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Outcomes of Hyperglycaemia in Pregnancy in Africa: Systematic Review Study Protocol

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Abstract

Introduction: The prevalence of diabetes mellitus globally has increased considerably over the past decades with a resultant increase in the incidence of diabetes complicated pregnancies. Hyperglycaemia in pregnancy is the most common metabolic complication encountered during pregnancy and is associated with adverse maternal and foetal outcomes. This systematic review aims to examine maternal, foetal, neonatal, childhood and long-term maternal outcomes of hyperglycaemia in pregnancy in Africa.

Methods and analysis: A systematic review of all studies that investigated hyperglycaemia in pregnancy outcomes, carried out in Africa from 1998 to 2019. A comprehensive search of all published articles indexed in Pubmed-MEDLINE, Cochrane library, Scopus, CINAHL via Ebscohost and Web of Science databases will be performed. Studies will be screened for eligibility by title, abstract and full text in duplicate by two independent reviewers. For data where meta-analysis is not possible, narrative analysis will be carried out using themes from data. For data where meta-analysis is possible, random-effects meta-analysis will be conducted. This systematic review will be reported according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses Protocol (PRISMA-P) 2015.

Ethics and dissemination: Ethical approval is not required for this study considering this is a systematic review protocol that utilises only published data. The findings of this study will be disseminated through peer-reviewed publications and conference presentations.

PROSPERO Registration Number: Awaiting registration number

Key Words

Diabetes mellitus, hyperglycaemia in pregnancy, hyperglycaemia first detected in pregnancy, gestational diabetes, Africa, maternal, child, pregnancy outcomes

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Strengths and limitations of this study

- The proposed study will provide updated knowledge on the prevalence of maternal, foetal, neonatal, childhood and long-term maternal and offspring outcomes of hyperglycaemia in pregnancy in Africa.

- The Burden of Disease Review Manager developed by the South African Medical Research Council will be utilised for extraction and recording of research data by two independent reviewers. The protocol adheres to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guideline

- Meta-analysis may not be possible for certain outcomes due to a limited number of eligible studies.

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Introduction

While the number of people with diabetes globally has considerably increased over the past few decades, so too has the incidence of diabetes complicated pregnancies.[1] Hyperglycaemia in pregnancy (HIP), defined by the WHO in 2014, includes diabetes first detected at any time during pregnancy, sub-classified as overt diabetes in pregnancy (DIP) and gestational diabetes (GDM) and pre-existing diabetes.[2] HIP is regarded as the most common metabolic complication encountered during pregnancy.[2] This is in part driven by the rising prevalence of type 2 diabetes (T2DM) and its risk factors and changing criteria for GDM.[3-6] The International Diabetes Federation (IDF) in 2013 reported that globally, 16.9% (21.4 million) of live births in women of reproductive age (16-49 years) were complicated by hyperglycaemia in pregnancy, with known or previously undiagnosed type 1 and 2 diabetes accounting for about 16%.[1]

Although the extent of the burden of HIP in Africa is not known, a recent systematic review reported that the pooled prevalence of T2DM in women of child-bearing age on the continent was 7.2%.[7] Further, a systematic review of the prevalence of GDM in Africa found a pooled prevalence of 13.6% with the prevalence in Sub-Saharan Africa reported at 9% and 14.8% by different groups.[8, 9]

HIP is associated with adverse maternal and foetal outcomes during pregnancy. These include increased caesarean delivery rate, preeclampsia, difficult labour, macrosomia, shoulder dystocia, perinatal mortality, neonatal hypoglycaemia and congenital malformations.[10-13] In addition, there are accumulating global data of the long-term impact of GDM on maternal health, childhood adiposity and glucose tolerance.[14-16]

There are isolated reports on pregnancy outcomes of women with pregestational diabetes in Africa.[17-19] In a systematic review of the burden, risk factors and outcomes of GDM pregnancies in SSA, Natamba et al found that up to 2018, 6 studies reported on pregnancy-related outcomes.[9] They found that GDM was associated with an increased risk of macrosomia (RR; 2.19, 95%CI: 1.08–4.43) and a non-significant risk of caesarean delivery..[9]

The aim of this systematic review is to examine the maternal, foetal, neonatal, childhood and long-term maternal and offspring outcomes of HIP in Africa. The data obtained will provide useful information for health systems planning and strengthening.

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The specific objectives are:

1. Primary - To estimate the prevalence of the following outcomes from HIP
    a. Maternal outcomes: preeclampsia, caesarean delivery rate
    b. Foetal/neonatal outcomes: foetal macrosomia, congenital anomalies, intrauterine foetal death, shoulder dystocia, neonatal morbidity (hypoglycaemia, sepsis, respiratory immaturity, jaundice, neonatal ICU admission and duration of neonatal hospital stay), perinatal mortality (early neonatal death (ENNDs) and stillbirth (SB) rates,

2. Secondary: To estimate the prevalence of the following outcomes from HIP:
    a. Maternal outcomes: miscarriage, preterm birth, antepartum and puerperal sepsis, gestational hypertension.
    b. Foetal/neonatal outcomes: birth trauma, small for gestational age (SGA) babies, infant deaths.
    c. Long-term maternal outcomes: Type 2 diabetes, metabolic syndrome, CVD risk factors
    d. Offspring outcomes: childhood overweight and obesity, and childhood pre-diabetes and diabetes as well as the prevalence of non-communicable diseases in adulthood (hypertension, diabetes, coronary heart disease, peripheral artery disease and cerebrovascular accidents)

Methods and analysis

Design

This research will utilise a systematic review of all literature published during the period 1998 – 2019. A meta-analysis will be carried out where sufficient data with low heterogeneity are available.

Inclusion criteria

Studies reporting the outcomes of HIP among women resident in Africa and published between 1998 to 2019 will be included as current criteria for the diagnosis of diabetes have been widely accepted since 1998. Participants will be included irrespective of their age, ethnicity, educational and socioeconomic status, gestational age, and study setting. Diagnosis of pre-gestational diabetes (type 1 or type 2 diabetes) and GDM will be defined according to WHO 1999/2013, ADA and IADPSG diagnostic criteria or definition.[20-23]

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All published and unpublished cohort or cross-sectional studies and baseline data from randomized controlled trials conducted in Africa reporting on the prevalence of the outcomes of HIP will be included.

Settings: Hospital and community-based studies.

Exclusion criteria:

a. Studies not performed in humans.
b. Studies not carried out in Africa
c. Studies in migrant African populations in regions other than Africa.
d. Reviews, commentaries, editorials, letters, and studies without primary data or explicit description of methods, or both.
e. Qualitative studies.

Definition of Outcomes

Definitions for key outcomes, where different criteria exist, are given below.

1. Preeclampsia defined as a multisystem disease that is characterized by new-onset hypertension that develops in the second half of pregnancy or after delivery in a woman not known to be hypertensive before and the coexistence of one or more of the following new-onset conditions: proteinuria, other maternal organ dysfunction (renal insufficiency, liver involvement, neurological complications and haematological complications) and uteroplacental dysfunction.[24]

2. Primary caesarean delivery defined as the delivery by caesarean section for the first time.[25]

3. Congenital malformations will be defined as any single or multiple defects of the morphogenesis of organs or body regions identifiable at birth or during the intrauterine life.[26]

4. Spontaneous abortion/miscarriage will be defined as any pregnancy loss before the 28th week of gestation (this is relevant for low and middle-income countries (LMIC) setting) or loss of foetus less than 500g.[27-29]

5. Perinatal mortality will be defined as a combination of stillbirths and early neonatal death (death before 7 days) (this is relevant for a LMIC setting).[28, 30]

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Search strategy for identification of relevant studies

A sensitive search strategy will be applied using the updated African search filter to retrieve published studies of outcomes of diabetes in pregnancy Search articles published in English and indexed in Pubmed-MEDLINE, Cochrane library, Scopus, CINAHL via Ebscohost and Web of Science databases will be retrieved from the start of each database until a present date to capture all relevant studies. Our search strategy will utilise medical subject heading (MeSH) and free text. Unpublished literature will be sought from experts in the field while grey literature such as reports will also be reviewed for relevant information from other organizational websites such as WHO, IDF, Google Scholar, and Pan African Clinical Trials Registry (PACTR). The support of an experienced Librarian will be sought for the validation and cross-examination of our search strategy

Study selection for this review

Two reviewers (EM, TC) will independently screen titles and abstracts of articles to identify eligible studies. Any discrepancies and doubts will be addressed via group discussion and consultation among the two reviewers (EM, TC).

Disagreements will be resolved through discussion and when needed there will be arbitration by a third reviewer (MW). Copies of eligible full texts will be retrieved and analysed by the 2 reviewers (EM, TC). Where necessary, further information will be sought from the authors of studies. Reasons for excluding articles will be recorded.

Data extraction

The Burden of Disease (BOD) Review Manager developed by the South African Medical Research Council will be utilised for extraction and recording of research data by two independent reviewers (EM, TC).[31] The data to be extracted include study details- publication date, title of study, study design, period and objective of the study; study population- country of study, study setting, and sample size; response rate, case definition of outcomes reported in the study, and study population characteristics. After the data extraction is completed, the 2 reviewers will address any identified difference or consult a third reviewer (MW). Where there is missing information, the corresponding author of the study will be contacted to request the missing information. No response to a maximum of three emails over 2 weeks sent to the corresponding author to request for additional information will lead to exclusion of the study. For studies appearing in more than one published article, we will consider the most recent,

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comprehensive, and with the largest sample size. For surveys appearing in one article with multiple surveys conducted at different time points, we shall treat each survey as a separate study. For multinational studies, data will be separated to show the estimate at country level.

Risk of bias assessment

Risk of bias and quality of study will be identified independently by two reviewers utilising a checklist for observational studies adapted from the risk of bias tool for population-based studies and the Newcastle-Ottawa Scale for assessing the quality of non-randomised studies with standardisation in the BOD review manager.[32, 33] External and internal validity will be assessed with discrepancies among reviewers over the risk of bias resolved by consensus and an additional reviewer when the need arises.

Data synthesis and heterogeneity assessment

Crude numerators and denominators from the individual studies will be used to re-calculate the study-specific prevalence. Prevalence estimates will be summarized by geographic regions and by outcomes. A meta-analysis will be performed on variables that are similar across the included studies.

For the meta-analysis, proportions will be stabilized using the double arcsine transformation[34], and then, a random-effects meta-analysis will be performed [35] to determine the pooled estimate of the prevalences. Heterogeneity will be explored using Cochrane's Q and quantified by I2 statistics.[36] Subgroup analyses will be performed based on the following: patient characteristics (age categories, sex, education level, socioeconomic status). The presence of publication bias will be assessed using Egger's test and funnel plots.[37] P-value < 0.10 on the Egger's test will be considered to be statistically significant for publication bias.[37] Inter-rater agreements between the researchers involved in study selection and those involved in the identification of risk of bias will be assessed using κ Cohen's coefficient.[38] All analyses will be performed using "Metaprop" routine using Stata version 15.[39] Results will be reported as proportions with corresponding 95% confidence intervals (CIs).

Review reporting

Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) 2015 statement will be used for reporting this systematic review.[40] The guideline for reporting systematic reviews of healthcare intervention will constitute a PRISMA checklist.

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Patient and public involvement: As no participant recruitment will be necessary, patients will not be involved in the design of this study protocol.

Ethics and Dissemination: Ethical approval will not be required for this study, as this is a systematic review protocol that utilises only published data and the study findings will be shared with the public either through peer-review publication or abstract presentation at conferences and possible submission to the relevant regional/global health policymaking bodies.

Authors' contributions: EM and NL designed the study. EM wrote the first manuscript of the review. EM, TC, MW, MM, SN and NL revised the review.
All authors read and approved the final manuscript.

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Competing interests: None declared

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a

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Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	9
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	9
Sponsor	#5b	Provide name for the review funder and / or sponsor	n/a
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	4

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		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	7

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process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	5, 6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5, 6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	8
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	8
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	8
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	8
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	8

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		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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Outcomes of Hyperglycaemia in Pregnancy in Africa: Systematic Review Study Protocol

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Abstract

Introduction: The prevalence of diabetes mellitus globally has increased considerably over the past decades with a resultant increase in the incidence of diabetes complicated pregnancies. Hyperglycaemia in pregnancy is the most common metabolic complication encountered during pregnancy and is associated with adverse maternal and foetal outcomes. This systematic review aims to examine maternal, foetal, neonatal, childhood and long-term maternal outcomes of hyperglycaemia in pregnancy in Africa.

Methods and analysis: A systematic review of all studies that investigated hyperglycaemia in pregnancy outcomes, carried out in Africa from 1998 to 2019. A comprehensive search of all published articles indexed in PubMed-MEDLINE, Cochrane library, Scopus, CINAHL (Ebscohost), Embase and Web of Science databases will be performed. Studies will be screened for eligibility by title, abstract and full text in duplicate by two independent reviewers. For data where meta-analysis is not possible, narrative analysis will be carried out using themes from data. For data where meta-analysis is possible, random-effects meta-analysis will be conducted. This systematic review will be reported according to the Meta-analyses Of Observational Studies in Epidemiology (MOOSE).

Ethics and dissemination: Ethical approval is not required for this study considering this is a systematic review protocol that utilises only published data. The findings of this study will be disseminated through peer-reviewed publications and conference presentations.

PROSPERO Registration Number: CRD42020184573

Key Words

Diabetes mellitus, hyperglycaemia in pregnancy, hyperglycaemia first detected in pregnancy, gestational diabetes, Africa, maternal, child, pregnancy outcomes

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Strengths and limitations of this study

- The proposed study will provide updated knowledge on the prevalence of maternal, foetal, neonatal, childhood and long-term maternal and offspring outcomes of hyperglycaemia in pregnancy in Africa.

- The Burden of Disease Review Manager developed by the South African Medical Research Council will be utilised for extraction and recording of research data by two independent reviewers. The protocol adheres to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guideline

- Meta-analysis may not be possible for certain outcomes due to a limited number of eligible studies.

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Introduction

While the number of people with diabetes globally has considerably increased over the past few decades, so too has the incidence of diabetes complicated pregnancies.[1] Hyperglycaemia in pregnancy (HIP), defined by the WHO in 2014, includes diabetes first detected at any time during pregnancy, sub-classified as overt diabetes in pregnancy (DIP) and gestational diabetes (GDM) and pre-existing diabetes.[2] HIP is regarded as the most common metabolic complication encountered during pregnancy.[2] This is in part driven by the rising prevalence of type 2 diabetes (T2DM) and its risk factors and changing criteria for GDM.[3-6] The International Diabetes Federation (IDF) in 2013 reported that globally, 16.9% (21.4 million) of live births in women of reproductive age (16-49 years) were complicated by hyperglycaemia in pregnancy, with known or previously undiagnosed type 1 and 2 diabetes accounting for about 16%.[1]

Although the extent of the burden of HIP in Africa is not known, a recent systematic review reported that the pooled prevalence of T2DM in women of child-bearing age on the continent was 7.2%.[7] Further, a systematic review of the prevalence of GDM in Africa found a pooled prevalence of 13.6% with the prevalence in Sub-Saharan Africa reported at 9% and 14.8% by different groups.[8, 9]

HIP is associated with adverse maternal and foetal outcomes during pregnancy. These include increased caesarean delivery rate, preeclampsia, difficult labour, macrosomia, shoulder dystocia, perinatal mortality, neonatal hypoglycaemia and congenital malformations.[10-13] In addition, there are accumulating global data of the long-term impact of GDM on maternal health, childhood adiposity and glucose tolerance.[14-16]

There are isolated reports on pregnancy outcomes of women with pregestational diabetes in Africa.[17-19] In a systematic review of the burden, risk factors and outcomes of GDM pregnancies in SSA, Natamba et al found that up to 2018, 6 studies reported on pregnancy-related outcomes.[9] They found that GDM was associated with an increased risk of macrosomia (RR; 2.19, 95%CI: 1.08–4.43) and a non-significant risk of caesarean delivery.[9]

The aim of this systematic review is to examine the maternal, foetal, neonatal, childhood and long-term maternal and offspring outcomes of HIP in Africa, a continent where many countries are experiencing an increasing prevalence of non-communicable diseases (NCDs), while burdens from infectious diseases continue to be high and health systems remain weak. The data obtained will provide useful information for health systems planning and strengthening.

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The specific objectives are:

1. Primary - To estimate the prevalence of the following outcomes from HIP in Africa
    a. Maternal outcomes: preeclampsia, caesarean delivery rate
    b. Foetal/neonatal outcomes: foetal macrosomia, congenital anomalies, intrauterine foetal death, shoulder dystocia, neonatal morbidity (hypoglycaemia, sepsis, respiratory immaturity, jaundice, neonatal ICU admission and duration of neonatal hospital stay), perinatal mortality (early neonatal death (ENNDs) and stillbirth (SB) rates,

2. Secondary: To estimate the prevalence of the following outcomes from HIP in Africa:
    a. Maternal outcomes: miscarriage, preterm birth, antepartum and puerperal sepsis, gestational hypertension.
    b. Foetal/neonatal outcomes: birth trauma, small for gestational age (SGA) babies, infant deaths.
    c. Long-term maternal outcomes: Type 2 diabetes, metabolic syndrome, CVD risk factors
    d. Offspring outcomes: childhood overweight and obesity, and childhood pre-diabetes and diabetes as well as the prevalence of non-communicable diseases in adulthood (hypertension, diabetes, coronary heart disease, peripheral artery disease and cerebrovascular accidents)

Methods and analysis

Design

This research will utilise a systematic review of all literature published during the period 1998 – 2019. A meta-analysis will be carried out where sufficient data with low heterogeneity are available.

Inclusion criteria

Studies reporting the outcomes of HIP among women resident in Africa and published between 1998 to 2019 will be included as current criteria for the diagnosis of diabetes have been widely accepted since 1998. Participants will be included irrespective of their age, ethnicity, educational and socioeconomic status, gestational age, and study setting. Diagnosis of pre-gestational diabetes (type 1 or type 2 diabetes) and GDM will be defined according to WHO 1999/2013, ADA and IADPSG diagnostic criteria or definition.[20-23]

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All published and unpublished cohort or cross-sectional studies and baseline data from randomized controlled trials conducted in Africa reporting on the prevalence of the outcomes of HIP will be included. Also, published multicentre studies where African patients were involved will be included.

Settings: Hospital and community-based studies.

Exclusion criteria:

a. Studies not performed in humans.
b. Studies not carried out in Africa
c. Studies in migrant African populations in regions other than Africa.
d. Reviews, commentaries, editorials, letters, and studies without primary data or explicit description of methods, or both.
e. Qualitative studies.

Definition of Outcomes

Definitions for key outcomes, where different criteria exist, are given below.

1. Preeclampsia defined as a multisystem progressive disorder characterised by the new onset of hypertension and proteinuria or the new onset of hypertension and significant end-organ dysfunction with or without proteinuria in the last half of pregnancy or postpartum.[24] The end-evidence of organ dysfunction include: i. proteinuria ≥0.3 g in a 24-hour urine specimen or protein/creatinine ratio ≥0.3 (mg/mg) (30 mg/mmol) in a random urine specimen or dipstick ≥2+ if a quantitative measurement is unavailable, ii. renal insufficiency- serum creatinine >1.1 mg/dL (97.2 µmol/L) or doubling of the creatinine concentration in the absence of other renal disease, iii. liver involvement- liver transaminases at least twice the upper limit of the normal concentrations for the local laboratory, iv. neurological complications- new-onset and persistent headache not accounted by alternative diagnoses and not responding to usual doses of analgesics, and visual symptoms (blurred vision, flashing lights or sparks, scotomata), and v. thrombocytopaenia- platelet count <100,000/µL.[24, 25]

2. Primary caesarean delivery defined as the delivery by caesarean section for the first time.[26]

3. Congenital malformations will be defined as any single or multiple defects of the morphogenesis of organs or body regions identifiable at birth or during the intrauterine life.[27]

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4. Spontaneous abortion/miscarriage will be defined as any pregnancy loss before the 28th week of gestation (this is relevant for low and middle-income countries (LMIC) setting) or loss of foetus less than 500g.[28-30]

5. Perinatal mortality will be defined as a combination of stillbirths and early neonatal death (death before 7 days) (this is relevant for a LMIC setting).[29, 31]

Search strategy for identification of relevant studies

A sensitive search strategy will be applied using the updated African search filter to retrieve published studies of outcomes of diabetes in pregnancy. Search articles published in English and indexed in PubMed-MEDLINE, Cochrane library, Scopus, CINAHL (Ebscohost), Embase and Web of Science databases will be retrieved from the start of each database until a present date to capture all relevant studies. Our search strategy will utilise medical subject heading (MeSH) and free text. Unpublished literature will be sought from experts in the field while grey literature such as reports will also be reviewed for relevant information from other organizational websites such as WHO, IDF, Google Scholar, and Pan African Clinical Trials Registry (PACTR). The support of an experienced Librarian will be sought for the validation and cross-examination of our search strategy

Study selection for this review

Two reviewers (EM, TC) will independently screen titles and abstracts of articles to identify eligible studies. Any discrepancies and doubts will be addressed via group discussion and consultation among the two reviewers (EM, TC).

Disagreements will be resolved through discussion and when needed there will be arbitration by a third reviewer (MW). Copies of eligible full texts will be retrieved and analysed by the 2 reviewers (EM, TC). Where necessary, further information will be sought from the authors of studies. Reasons for excluding articles will be recorded.

Data extraction

The Burden of Disease (BOD) Review Manager developed by the South African Medical Research Council will be utilised for extraction and recording of research data by two independent reviewers (EM, TC).[32] The data to be extracted include study details- publication date, title of study, study design, period and objective of the study; study population- country of study, study setting, and sample size; response rate, case definition of outcomes reported in

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the study, and study population characteristics. After the data extraction is completed, the 2 reviewers will address any identified difference or consult a third reviewer (MW). Where there is missing information, the corresponding author of the study will be contacted to request the missing information. No response to a maximum of three emails over 2 weeks sent to the corresponding author to request for additional information will lead to exclusion of the study. For studies appearing in more than one published article, we will consider the most recent, comprehensive, and with the largest sample size. For surveys appearing in one article with multiple surveys conducted at different time points, we shall treat each survey as a separate study. For multinational studies, data will be separated to show the estimate at country level.

Risk of bias assessment

Risk of bias and quality of study will be identified independently by two reviewers utilising a checklist for observational studies adapted from the risk of bias tool for population-based studies and the Newcastle-Ottawa Scale for assessing the quality of non-randomised studies with standardisation in the BOD review manager, while Cochrane Risk of Bias tool will be used for Randomised Controlled Trials (RCTs)Ts.[33, 34] External and internal validity will be assessed with discrepancies among reviewers over the risk of bias resolved by consensus and an additional reviewer when the need arises.

Data synthesis and heterogeneity assessment

Crude numerators and denominators from the individual studies will be used to re-calculate the study-specific prevalence. Prevalence estimates will be summarized by geographic regions and by outcomes. A meta-analysis will be performed on variables that are similar across the included studies.

For the meta-analysis, proportions will be stabilized using the double arcsine transformation[35], and then, a random-effects meta-analysis will be performed [36] to determine the pooled estimate of the prevalences. Heterogeneity will be explored using Cochrane's Q and quantified by I2 statistics.[37] Subgroup analyses will be performed based on the following: patient characteristics (age categories, sex, education level, socioeconomic status). The presence of publication bias will be assessed using Egger's test and funnel plots.[38] P-value < 0.10 on the Egger's test will be considered to be statistically significant for publication bias.[38] Inter-rater agreements between the researchers involved in study selection and those involved in the identification of risk of bias will be assessed using κ Cohen's coefficient.[39] All analyses will be performed using

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"Metaprop" routine using Stata version 15.[40] Results will be reported as proportions with corresponding 95% confidence intervals (CIs).

Review reporting

Meta-analyses Of Observational Studies in Epidemiology (MOOSE) will be used for reporting this systematic review.[41] The guideline for reporting systematic reviews of healthcare intervention will constitute a MOOSE checklist.

Patient and public involvement: As no participant recruitment will be necessary, patients will not be involved in the design of this study protocol.

Ethics and Dissemination: Ethical approval will not be required for this study, as this is a systematic review protocol that utilises only published data and the study findings will be shared with the public either through peer-review publication or abstract presentation at conferences and possible submission to the relevant regional/global health policymaking bodies.

Authors' contributions: EM and NL designed the study. EM wrote the first manuscript of the review. EM, TC, MW, MM, SN and NL revised the review.

All authors read and approved the final manuscript.

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Problem definition		
Hypothesis statement		
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Type of exposure or intervention used		
Type of study design used		
Study population		
Reporting of Search Strategy		
Qualifications of searchers (eg, librarians and investigators)		
Search strategy, including time period included in the synthesis and keywords		
Effort to include all available studies, including contact with authors		
Databases and registries searched		
Search software used, name and version, including special features used (eg, explosion)		
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Outcomes of Hyperglycaemia in Pregnancy in Africa: Systematic Review Study Protocol

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Abstract

Introduction: The prevalence of diabetes mellitus globally has increased considerably over the past decades with a resultant increase in the incidence of diabetes complicated pregnancies. Hyperglycaemia in pregnancy is the most common metabolic complication encountered during pregnancy and is associated with adverse maternal and foetal outcomes. This systematic review aims to examine maternal, foetal, neonatal, childhood and long-term maternal outcomes of hyperglycaemia in pregnancy in Africa.

Methods and analysis: A systematic review of all studies that investigated hyperglycaemia in pregnancy outcomes, carried out in Africa from 1998 to 2019. A comprehensive search of all published articles indexed in PubMed-MEDLINE, Cochrane library, Scopus, CINAHL (Ebscohost), Embase and Web of Science databases will be performed. Studies will be screened for eligibility by title, abstract and full text in duplicate by two independent reviewers. For data where meta-analysis is not possible, narrative analysis will be carried out using themes from data. For data where meta-analysis is possible, random-effects meta-analysis will be conducted. This systematic review will be reported according to the Meta-analyses Of Observational Studies in Epidemiology (MOOSE).

Ethics and dissemination: Ethical approval is not required for this study considering this is a systematic review protocol that utilises only published data. The findings of this study will be disseminated through peer-reviewed publications and conference presentations.

PROSPERO Registration Number: CRD42020184573

Key Words

Diabetes mellitus, hyperglycaemia in pregnancy, hyperglycaemia first detected in pregnancy, gestational diabetes, Africa, maternal, child, pregnancy outcomes

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Strengths and limitations of this study

- The proposed study will provide updated knowledge on the prevalence of maternal, foetal, neonatal, childhood and long-term maternal and offspring outcomes of hyperglycaemia in pregnancy in Africa.

- The Burden of Disease Review Manager developed by the South African Medical Research Council will be utilised for extraction and recording of research data by two independent reviewers.

- The protocol adheres to the Meta-analyses Of Observational Studies in Epidemiology reporting guideline.

- Meta-analysis may not be possible for certain outcomes due to a limited number of eligible studies.

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Introduction

While the number of people with diabetes globally has considerably increased over the past few decades, so too has the incidence of diabetes complicated pregnancies.[1] Hyperglycaemia in pregnancy (HIP), defined by the WHO in 2014, includes diabetes first detected at any time during pregnancy, sub-classified as overt diabetes in pregnancy (DIP) and gestational diabetes (GDM) and pre-existing diabetes.[2] HIP is regarded as the most common metabolic complication encountered during pregnancy.[2] This is in part driven by the rising prevalence of type 2 diabetes (T2DM) and its risk factors and changing criteria for GDM.[3-6] The International Diabetes Federation (IDF) in 2013 reported that globally, 16.9% (21.4 million) of live births in women of reproductive age (16-49 years) were complicated by hyperglycaemia in pregnancy, with known or previously undiagnosed type 1 and 2 diabetes accounting for about 16%.[1]

Although the extent of the burden of HIP in Africa is not known, a recent systematic review reported that the pooled prevalence of T2DM in women of child-bearing age on the continent was 7.2%.[7] Further, a systematic review of the prevalence of GDM in Africa found a pooled prevalence of 13.6% with the prevalence in Sub-Saharan Africa reported at 9% and 14.8% by different groups.[8, 9]

HIP is associated with adverse maternal and foetal outcomes during pregnancy. These include increased caesarean delivery rate, preeclampsia, difficult labour, macrosomia, shoulder dystocia, perinatal mortality, neonatal hypoglycaemia and congenital malformations.[10-13] In addition, there are accumulating global data of the long-term impact of GDM on maternal health, childhood adiposity and glucose tolerance.[14-16]

There are isolated reports on pregnancy outcomes of women with pregestational diabetes in Africa.[17-19] In a systematic review of the burden, risk factors and outcomes of GDM pregnancies in SSA, Natamba et al found that up to 2018, 6 studies reported on pregnancy-related outcomes.[9] They found that GDM was associated with an increased risk of macrosomia (RR; 2.19, 95%CI: 1.08–4.43) and a non-significant risk of caesarean delivery.[9]

The aim of this systematic review is to examine the maternal, foetal, neonatal, childhood and long-term maternal and offspring outcomes of HIP in Africa, a continent where many countries are experiencing an increasing prevalence of non-communicable diseases (NCDs), while burdens from infectious diseases continue to be high and health systems remain weak. The data obtained will provide useful information for health systems planning and strengthening.

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The specific objectives are:

1. Primary - To estimate the prevalence of the following outcomes from HIP in Africa
    a. Maternal outcomes: preeclampsia, caesarean delivery rate
    b. Foetal/neonatal outcomes: foetal macrosomia, congenital anomalies, intrauterine foetal death, shoulder dystocia, neonatal morbidity (hypoglycaemia, sepsis, respiratory immaturity, jaundice, neonatal ICU admission and duration of neonatal hospital stay), perinatal mortality (early neonatal death (ENNDs) and stillbirth (SB) rates,

2. Secondary: To estimate the prevalence of the following outcomes from HIP in Africa:
    a. Maternal outcomes: miscarriage, preterm birth, antepartum and puerperal sepsis, gestational hypertension.
    b. Foetal/neonatal outcomes: birth trauma, small for gestational age (SGA) babies, infant deaths.
    c. Long-term maternal outcomes: Type 2 diabetes, metabolic syndrome, CVD risk factors
    d. Offspring outcomes: childhood overweight and obesity, and childhood pre-diabetes and diabetes as well as the prevalence of non-communicable diseases in adulthood (hypertension, diabetes, coronary heart disease, peripheral artery disease and cerebrovascular accidents)

Methods and analysis

Design

This research will utilise a systematic review of all literature published during the period 1998 – 2019. A meta-analysis will be carried out where sufficient data with low heterogeneity are available.

Inclusion criteria

Studies reporting the outcomes of HIP among women resident in Africa and published between 1998 to 2019 will be included as current criteria for the diagnosis of diabetes have been widely accepted since 1998. Participants will be included irrespective of their age, ethnicity, educational and socioeconomic status, gestational age, and study setting. Diagnosis of pre-gestational diabetes (type 1 or type 2 diabetes) and GDM will be defined according to WHO 1999/2013, ADA and IADPSG diagnostic criteria or definition.[20-23]

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All published and unpublished cohort or cross-sectional studies and baseline data from randomized controlled trials conducted in Africa reporting on the prevalence of the outcomes of HIP will be included. Also, published multicentre studies where African patients were involved will be included.

Settings: Hospital and community-based studies.

Exclusion criteria:

a. Studies not performed in humans.
b. Studies not carried out in Africa
c. Studies in migrant African populations in regions other than Africa.
d. Reviews, commentaries, editorials, letters, and studies without primary data or explicit description of methods, or both.
e. Qualitative studies.

Definition of Outcomes

Definitions for key outcomes, where different criteria exist, are given below.

1. Preeclampsia defined as a multisystem progressive disorder characterised by the new onset of hypertension and proteinuria or the new onset of hypertension and significant end-organ dysfunction with or without proteinuria in the last half of pregnancy or postpartum.[24] The end-evidence of organ dysfunction include: i. proteinuria ≥0.3 g in a 24-hour urine specimen or protein/creatinine ratio ≥0.3 (mg/mg) (30 mg/mmol) in a random urine specimen or dipstick ≥2+ if a quantitative measurement is unavailable, ii. renal insufficiency- serum creatinine >1.1 mg/dL (97.2 µmol/L) or doubling of the creatinine concentration in the absence of other renal disease, iii. liver involvement- liver transaminases at least twice the upper limit of the normal concentrations for the local laboratory, iv. neurological complications- new-onset and persistent headache not accounted by alternative diagnoses and not responding to usual doses of analgesics, and visual symptoms (blurred vision, flashing lights or sparks, scotomata), and v. thrombocytopaenia- platelet count <100,000/µL.[24, 25]

2. Primary caesarean delivery defined as the delivery by caesarean section for the first time.[26]

3. Congenital malformations will be defined as any single or multiple defects of the morphogenesis of organs or body regions identifiable at birth or during the intrauterine life.[27]

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4. Spontaneous abortion/miscarriage will be defined as any pregnancy loss before the 28th week of gestation (this is relevant for low and middle-income countries (LMIC) setting) or loss of foetus less than 500g.[28-30]

5. Perinatal mortality will be defined as a combination of stillbirths and early neonatal death (death before 7 days) (this is relevant for a LMIC setting).[29, 31]

Patient and public involvement: As no participant recruitment will be necessary, patients will not be involved in the design of this study protocol.

Search strategy for identification of relevant studies

A sensitive search strategy will be applied using the updated African search filter to retrieve published studies of outcomes of diabetes in pregnancy. Search articles published in English and indexed in PubMed-MEDLINE, Cochrane library, Scopus, CINAHL (Ebscohost), Embase and Web of Science databases will be retrieved from the start of each database until a present date to capture all relevant studies. Our search strategy will utilise medical subject heading (MeSH) and free text. Unpublished literature will be sought from experts in the field while grey literature such as reports will also be reviewed for relevant information from other organizational websites such as WHO, IDF, Google Scholar, and Pan African Clinical Trials Registry (PACTR). The support of an experienced Librarian will be sought for the validation and cross-examination of our search strategy

Study selection for this review

Two reviewers (EM, TC) will independently screen titles and abstracts of articles to identify eligible studies. Any discrepancies and doubts will be addressed via group discussion and consultation among the two reviewers (EM, TC).

Disagreements will be resolved through discussion and when needed there will be arbitration by a third reviewer (MW). Copies of eligible full texts will be retrieved and analysed by the 2 reviewers (EM, TC). Where necessary, further information will be sought from the authors of studies. Reasons for excluding articles will be recorded.

Data extraction

The Burden of Disease (BOD) Review Manager developed by the South African Medical Research Council will be utilised for extraction and recording of research data by two independent reviewers (EM, TC).[32] The data to be extracted include study details- publication

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date, title of study, study design, period and objective of the study; study population- country of study, study setting, and sample size; response rate, case definition of outcomes reported in the study, and study population characteristics. After the data extraction is completed, the 2 reviewers will address any identified difference or consult a third reviewer (MW). Where there is missing information, the corresponding author of the study will be contacted to request the missing information. No response to a maximum of three emails over 2 weeks sent to the corresponding author to request for additional information will lead to exclusion of the study. For studies appearing in more than one published article, we will consider the most recent, comprehensive, and with the largest sample size. For surveys appearing in one article with multiple surveys conducted at different time points, we shall treat each survey as a separate study. For multinational studies, data will be separated to show the estimate at country level.

Risk of bias assessment

Risk of bias and quality of study will be identified independently by two reviewers utilising a checklist for observational studies adapted from the risk of bias tool for population-based studies and the Newcastle-Ottawa Scale for assessing the quality of non-randomised studies with standardisation in the BOD review manager, while Cochrane Risk of Bias tool will be used for Randomised Controlled Trials (RCTs)Ts.[33, 34] External and internal validity will be assessed with discrepancies among reviewers over the risk of bias resolved by consensus and an additional reviewer when the need arises.

Data synthesis and heterogeneity assessment

Crude numerators and denominators from the individual studies will be used to re-calculate the study-specific prevalence. Prevalence estimates will be summarized by geographic regions and by outcomes. A meta-analysis will be performed on variables that are similar across the included studies.

For the meta-analysis, proportions will be stabilized using the double arcsine transformation[35], and then, a random-effects meta-analysis will be performed [36] to determine the pooled estimate of the prevalences. Heterogeneity will be explored using Cochrane's Q and quantified by I2 statistics.[37] Subgroup analyses will be performed based on the following: patient characteristics (age categories, sex, education level, socioeconomic status). The presence of publication bias will be assessed using Egger's test and funnel plots.[38] P-value < 0.10 on the Egger's test will be considered to be statistically significant for publication bias.[38] Inter-rater agreements between the researchers involved in study selection and those involved in the identification of

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risk of bias will be assessed using κ Cohen's coefficient.[39] All analyses will be performed using "Metaprop" routine using Stata version 15.[40] Results will be reported as proportions with corresponding 95% confidence intervals (CIs).

Review reporting

Meta-analyses Of Observational Studies in Epidemiology (MOOSE) will be used for reporting this systematic review.[41] The guideline for reporting systematic reviews of healthcare intervention will constitute a MOOSE checklist.

Ethics and Dissemination: Ethical approval will not be required for this study, as this is a systematic review protocol that utilises only published data and the study findings will be shared with the public either through peer-review publication or abstract presentation at conferences and possible submission to the relevant regional/global health policymaking bodies.

Authors' contributions: EM and NL designed the study. EM wrote the first manuscript of the review. EM, TC, MW, MM, SN and NL revised the review.

All authors read and approved the final manuscript.

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Competing interests: None declared

Word Count: 1982

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MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	4
2	Hypothesis statement	-
3	Description of study outcome(s)	6-7
4	Type of exposure or intervention used	N/A
5	Type of study designs used	6
6	Study population	5
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	7
8	Search strategy, including time period included in the synthesis and key words	7
9	Effort to include all available studies, including contact with authors	5-6
10	Databases and registries searched	7
11	Search software used, name and version, including special features used (eg, explosion)	N/A
12	Use of hand searching (eg, reference lists of obtained articles)	N/A
13	List of citations located and those excluded, including justification	N/A
14	Method of addressing articles published in languages other than English	-
15	Method of handling abstracts and unpublished studies	7
16	Description of any contact with authors	7
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	7
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	7
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	N/A
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	N/A
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	8
22	Assessment of heterogeneity	8
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	8-9
24	Provision of appropriate tables and graphics	N/A
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	N/A
26	Table giving descriptive information for each study included	N/A
27	Results of sensitivity testing (eg, subgroup analysis)	N/A
28	Indication of statistical uncertainty of findings	N/A

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Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	N/A
30	Justification for exclusion (eg, exclusion of non-English language citations)	6
31	Assessment of quality of included studies	N/A
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	N/A
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	N/A
34	Guidelines for future research	-
35	Disclosure of funding source	N/A

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

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